Alternating Hard-Motif MoS-Poisson-SuSiE

1 Purpose

This note describes the current implemented mixture-of-submanifolds Poisson-SuSiE algorithm. It is an intermediate model between the MF-Poisson-SuSiE model and the final soft submanifold model. The current version uses hard submanifold motifs and alternates between:

1. fitting observation-level mixture responsibilities,
2. updating the shared factor matrix F, and
3. updating hard submanifold motifs.

The hard motifs make the algorithm easy to debug and provide a concrete test of whether a submanifold layer can improve the tree-like sparsity scenarios.

2 Model

We observe a count matrix $Y \in \mathbb{N}^{N\times M}$. Let $F \in \mathbb{R}_+^{K\times M}$ be a row-normalized factor matrix:

$$F_{km} \geq 0, \qquad \sum_{m=1}^{M} F_{km} = 1, \quad k \in [K].$$

There are S submanifolds. Each submanifold $s \in [S]$ has D hard motif slots,

$$a_{sd} \in [K], \qquad d \in [D],$$

where a_{sd} is the factor used by slot d of submanifold s. Repeated entries are allowed. Thus a submanifold with effective dimension one may be represented as $(k, k, \dots, k)$.

For observation i, let $z_i \in [S]$ denote its submanifold assignment. The current hard-motif generative model is:

$$z_i \sim \mathrm{Multi}(1, \pi), \tag{1}$$

$$\beta_{isd} \mid z_i = s \sim \mathrm{Gam}(\alpha^0_{sd}, \lambda^0_{sd}), \qquad d \in [D], \tag{2}$$

$$y_{im} \mid z_i = s, \beta_{is1:D}, F, a \sim \mathrm{Poi}\left(\sum_{d=1}^{D} \beta_{isd} F_{a_{sd},m}\right), \qquad m \in [M]. \tag{3}$$

This can be viewed as a mixture model over sparse Poisson factorization submanifolds. Conditional on $z_i = s$, the observation can only use factors listed in the hard motif $a_{s1:D}$.

3 Variational family

The current implementation uses:

$$q(z, \beta) = \prod_{i=1}^{N} q(z_i) \prod_{s=1}^{S} \prod_{d=1}^{D} q(\beta_{isd} \mid z_i = s),$$

where

$$q(z_i = s) = \omega_{is}, \qquad q(\beta_{isd} \mid z_i = s) = \mathrm{Gam}(\alpha_{isd}, \lambda_{isd}).$$

For each candidate submanifold s, auxiliary variables $\xi_{isdm} \geq 0$ are introduced with

$$\sum_{d=1}^{D} \xi_{isdm} = 1, \qquad i \in [N],\ s \in [S],\ m \in [M].$$

They allocate count y_{im} across the D motif slots of submanifold s.

4 Conditional component updates

Fix F, the hard motif matrix a, and prior parameters (α^0, λ^0). For each submanifold s, the conditional Poisson-SuSiE update has the same form as the fixed-motif single-observation Poisson update.

Auxiliary allocation. For each i, s, m, d,

$$\xi_{isdm} \leftarrow \frac{\exp\{\mathbb{E}_q[\log \beta_{isd}] + \log F_{a_{sd},m}\}}{\sum_{d'=1}^{D} \exp\{\mathbb{E}_q[\log \beta_{isd'}] + \log F_{a_{sd'},m}\}},$$

where

$$\mathbb{E}_q[\log \beta_{isd}] = \psi(\alpha_{isd}) - \log \lambda_{isd}.$$

Gamma posterior. For each i, s, d,

$$\alpha_{isd} \leftarrow \sum_{m=1}^{M} \xi_{isdm} y_{im} + \alpha^0_{sd}, \tag{4}$$

$$\lambda_{isd} \leftarrow \sum_{m=1}^{M} F_{a_{sd},m} + \lambda^0_{sd}. \tag{5}$$

Since F is row-normalized, $\sum_m F_{a_{sd},m} = 1$ in the current implementation.

Component lower bound. Let $\mathcal{L}_{is}$ denote the conditional variational lower bound for observation i under submanifold s. Up to constants not depending on s, the implemented quantity is

$$\begin{aligned}\mathcal{L}_{is} = &\sum_{d,m} \xi_{isdm} y_{im} \{\mathbb{E}_q[\log \beta_{isd}] + \log F_{a_{sd},m}\} - \sum_{d,m} y_{im} \xi_{isdm} \log \xi_{isdm} \\ &- \sum_{d} \mathbb{E}_q[\beta_{isd}] \sum_{m} F_{a_{sd},m} - \sum_{d} \mathrm{KL}\left\{\mathrm{Gam}(\alpha_{isd}, \lambda_{isd}) \,\|\, \mathrm{Gam}(\alpha^0_{sd}, \lambda^0_{sd})\right\}.\end{aligned} \tag{6}$$

5 Mixture responsibility update

Given component lower bounds $\mathcal{L}_{is}$, the observation-level responsibility update is

$$\omega_{is} \leftarrow \frac{\pi_s \exp(\mathcal{L}_{is})}{\sum_{s'=1}^{S} \pi_{s'} \exp(\mathcal{L}_{is'})}.$$

The mixing weights are updated by

$$\pi_s \leftarrow \frac{1}{N} \sum_{i=1}^{N} \omega_{is}.$$

6 Empirical-Bayes prior update

The current implementation optionally updates the motif-slot Gamma prior parameters. For each submanifold s and slot d, compute responsibility- weighted posterior moments:

$$\bar{\beta}_{sd} = \frac{\sum_i \omega_{is}\, \alpha_{isd}/\lambda_{isd}}{\sum_i \omega_{is}}, \tag{7}$$

$$\overline{\log \beta}_{sd} = \frac{\sum_i \omega_{is} \{\psi(\alpha_{isd}) - \log \lambda_{isd}\}}{\sum_i \omega_{is}}. \tag{8}$$

The Gamma shape solves

$$\log \alpha^0_{sd} - \psi(\alpha^0_{sd}) = \log \bar{\beta}_{sd} - \overline{\log \beta}_{sd},$$

using Newton's method, and then

$$\lambda^0_{sd} \leftarrow \frac{\alpha^0_{sd}}{\bar{\beta}_{sd}}.$$

7 Shared factor update

Given current responsibilities ω, auxiliary allocations ξ, and hard motifs a, the row-normalized F update collects responsibility-weighted expected counts:

$$A_{km} = \sum_{i=1}^{N} \sum_{s=1}^{S} \sum_{d=1}^{D} \omega_{is}\, \mathbf{1}\{a_{sd} = k\}\, \xi_{isdm}\, y_{im}.$$

Then

$$F^{\text{new}}_{km} \leftarrow \frac{A_{km} + \epsilon}{\sum_{m'=1}^{M} (A_{km'} + \epsilon)}.$$

The implementation supports a damped update

$$F_{k\cdot} \leftarrow (1 - \eta_t) F_{k\cdot} + \eta_t F^{\text{new}}_{k\cdot}, \qquad F_{k\cdot} \leftarrow \frac{F_{k\cdot}}{\sum_m F_{km}},$$

where η_t is ramped from an initial value to 1.

8 Hard motif update

The current hard motif update is intentionally simple. It first aggregates counts by submanifold responsibility:

$$Y_{sm}^{(S)} = \sum_{i=1}^{N} \omega_{is} y_{im}.$$

For fixed F, it then fits nonnegative submanifold loadings $H \in \mathbb{R}_{+}^{S \times K}$ by KL-NMF multiplicative updates:

$$H_{sk} \leftarrow H_{sk} \frac{\sum_m (Y_{sm}^{(S)}/(HF)_{sm}) F_{km}}{\sum_m F_{km}}.$$

The initialization of H uses the current hard motif counts, so the update is local around the current motif state.

After fitting H, define normalized submanifold loading shares

$$r_{sk} = \frac{H_{sk}}{\sum_{\ell=1}^{K} H_{s\ell}}.$$

For each submanifold s, the updated hard motif is constructed by selecting factors with r_{sk} above a threshold τ, keeping at most D factors, and padding with repeats if fewer than D factors are selected:

$$a_{s1:D} \leftarrow \mathrm{Pad}_D\left(\{k : r_{sk} \geq \tau\}\right).$$

This allows the fitted effective submanifold dimension to be smaller than the nominal D.

Remark 1 (Why this motif update is not the final model)**.** This update learns hard motifs, not posterior motif probabilities. It is useful as an intermediate coordinate step, but it does not yet quantify uncertainty over submanifold-level supports. The final model should replace the hard motif matrix a with soft submanifold-level posterior probabilities over factors.

9 Algorithm

Algorithm 1 Alternating hard-motif MoS-Poisson-SuSiE

1: Initialize F by running MF-Poisson-SuSiE with learned F.
2: Compute posterior loading scores from the MF-Poisson-SuSiE fit:

$$\widehat{L}_{ik} = \sum_{d=1}^{D} q(\gamma_{id} = k)\, \mathbb{E}_q[\beta_{id} \mid \gamma_{id} = k].$$

3: Cluster row-normalized $\widehat{L}$ into S groups.
4: Initialize hard motifs $a_{s1:D}$ from the cluster centers, allowing effective dimension smaller than D by thresholding and padding with repeats.
5: Initialize $\pi_s = 1/S$ and $\xi_{isdm} = 1/D$.
6: **for** $t = 1, \ldots, T$ **do**
7: **Conditional component update:** for every submanifold s, update ξ_{isdm}, α_{isd}, and λ_{isd} using the current F and a.
8: **Mixture update:** compute component lower bounds $\mathcal{L}_{is}$ and update responsibilities ω_{is}.
9: Update mixing proportions $\pi_s \leftarrow N^{-1} \sum_i \omega_{is}$.
10: **Prior update** (optional): update $(\alpha^0_{sd}, \lambda^0_{sd})$ using the responsibility-weighted Gamma moment equations.
11: **Factor update** (optional): compute A_{km} and update row-normalized F using a damped step.
12: **Motif update** (optional): aggregate $Y^{(S)}_{sm} = \sum_i \omega_{is} y_{im}$, fit submanifold loading shares r_{sk} with fixed-F KL-NMF updates, and update hard motifs $a_{s1:D}$ by thresholding and padding.
13: Record the mixture lower bound $\sum_i \log\{\sum_s \pi_s \exp(\mathcal{L}_{is})\}$.
14: **end for**
15: Refit the conditional components and responsibilities one final time using the returned F and motifs a.

10 Current interpretation

The current algorithm should be interpreted as a pragmatic hard-coordinate approximation. It answers the question:

> Can a submanifold mixture layer, initialized from MF-Poisson-SuSiE, improve recovery of repeated sparsity patterns and tree-like support structures?

The present implementation does learn F and hard motifs alternately, but only after an MF-Poisson-SuSiE initialization. The motif update is hard and local, and therefore the ELBO recorded before the F and motif updates is best treated as a diagnostic rather than as a guaranteed monotone objective. The next theoretical step is to derive and implement the soft motif posterior version of the model.